

NIKOLO-POLOMA, Russian Federation

WMO: 272520

Lat: 58.350N

Lon: 43.383E

Elev: 144

StdP: 99.60

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.8	-25.4	-31.6	0.2	-28.7	-28.0	0.3	-25.3	8.8	-6.9	7.9	-3.8	1.4	20	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	28.9	19.7	27.0	18.9	25.1	18.0	21.1	26.9	20.0	25.3	19.0	23.5	2.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.1	23.2	18.1	13.3	22.3	17.2	12.5	21.6	61.6	27.1	57.8	25.4	54.5	23.5	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	6.9	6.2	DB	-32.8	31.3	2.8	2.0	-34.8	32.7	-36.4	33.9	-38.0	35.0	-40.0	36.4	(4)
(5)			WB	-32.3	22.9	2.9	1.4	-34.4	23.9	-36.1	24.7	-37.7	25.5	-39.7	26.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-10.6	-9.7	-3.9	4.2	11.1	15.5	18.3	15.5	10.0	3.5	-3.3	-7.7
	DBStd	11.37	7.63	7.22	4.86	4.94	5.05	4.22	3.65	3.81	4.01	4.44	5.77	7.29
	HDD10.0	3061	638	551	431	185	49	5	1	2	48	206	398	548
	HDD18.3	5446	897	784	689	425	230	106	46	104	252	460	648	806
	CDD10.0	746	0	0	0	10	82	169	259	174	47	4	0	0
	CDD18.3	89	0	0	0	0	4	20	47	17	1	0	0	0
	CDH23.3	847	0	0	0	2	61	191	432	157	6	0	0	0
	CDH26.7	210	0	0	0	0	11	44	108	46	1	0	0	0
Wind	WSAvg	2.9	3.1	3.1	3.3	3.0	3.0	2.6	2.2	2.3	2.6	3.1	3.1	3.2
Precipitation	PrecAvg	629	37	28	31	36	49	87	71	71	56	65	47	41
	PrecMax	909	71	82	90	83	96	177	192	169	102	131	77	67
	PrecMin	483	15	5	3	15	21	22	3	26	19	24	9	16
	PrecStd	96	15	17	17	20	20	35	44	36	23	31	19	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	2.6	9.4	22.3	28.2	30.6	32.1	31.1	24.1	17.4	9.3	5.2
		MCWB	1.1	1.2	4.3	13.1	18.0	20.3	20.9	20.5	15.9	12.1	8.2	4.8
	2%	DB	0.8	1.3	5.7	18.8	24.9	27.7	29.4	26.9	21.1	13.6	6.3	2.7
		MCWB	0.4	0.5	2.2	10.8	15.9	19.0	20.0	19.4	15.4	10.2	5.5	2.1
	5%	DB	0.2	0.6	3.8	15.3	22.2	25.4	27.5	24.6	18.7	11.4	4.7	1.3
		MCWB	-0.2	0.0	1.1	8.9	14.1	17.7	19.4	18.0	14.2	9.4	4.0	0.8
	10%	DB	-0.7	-0.2	2.4	12.5	19.9	23.3	25.8	22.3	16.6	9.6	3.2	0.4
		MCWB	-1.1	-0.7	0.6	7.3	12.5	16.8	18.9	16.9	13.0	8.3	2.6	0.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.3	1.5	4.9	13.4	19.2	21.7	22.7	22.0	18.0	12.8	8.3	5.0
		MCDB	1.5	2.0	8.1	20.9	26.2	28.1	27.8	28.9	21.2	15.8	9.1	5.2
	2%	WB	0.5	0.7	2.9	11.5	17.1	20.3	21.4	20.2	16.2	11.0	5.8	2.3
		MCDB	0.7	1.2	4.6	17.6	23.2	25.7	27.2	25.5	19.6	13.1	6.1	2.6
	5%	WB	-0.1	0.1	1.8	9.7	15.4	18.9	20.4	19.0	14.8	9.6	4.1	0.9
		MCDB	0.2	0.6	3.1	14.3	20.1	23.7	25.8	22.9	17.9	11.0	4.5	1.2
	10%	WB	-1.2	-0.8	0.9	7.8	13.8	17.6	19.4	17.7	13.4	8.2	2.6	0.1
		MCDB	-0.8	-0.2	2.1	11.9	17.9	21.8	24.2	21.4	16.1	9.6	3.1	0.4
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.8	7.7	9.2	10.5	9.8	10.3	9.4	8.2	5.4	4.0	4.7
		MCDBR	3.9	4.7	8.5	14.3	14.0	13.2	13.2	12.6	11.7	8.9	4.9	3.5
		MCWBR	3.9	4.4	5.8	7.8	6.4	5.6	5.2	5.6	6.2	5.9	4.4	3.4
	5% WB	MCDBR	3.8	4.5	6.3	13.2	11.8	11.3	11.4	11.2	9.9	7.6	4.3	3.2
		MCWBR	3.9	4.4	4.6	7.7	6.1	5.3	5.3	5.5	5.6	5.6	4.3	3.3
Clear-Sky Solar Irradiance	taub	0.275	0.312	0.343	0.395	0.369	0.366	0.401	0.401	0.359	0.339	0.303	0.263	
	taud	2.067	2.062	2.059	2.134	2.393	2.423	2.341	2.318	2.420	2.355	2.184	2.039	
	Ebn at Noon	529	686	779	804	864	871	826	796	778	673	499	411	
	Edh at Noon	53	87	117	129	107	105	112	107	83	66	48	39	
All-Sky Solar Radiation	RadAvg	0.41	1.15	2.53	3.84	5.18	5.43	5.41	4.11	2.41	1.03	0.38	0.23	
	RadStd	0.07	0.28	0.32	0.49	0.41	0.45	0.50	0.33	0.32	0.19	0.06	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.82	N/A	N/A	N/A	N/A	N/A	N/A	-235	N/A	N/A
Regional (6 neighbors)		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	-208	N/A	N/A

Nomenclature: See separate page